

Cancer Data Science

Thursday, July 24th, 2025



Background: Breast Cancer

- Very prevalent cancer
- High burden of disease
- Patients can experience recurrence or death
- Many variables can impact outcomes



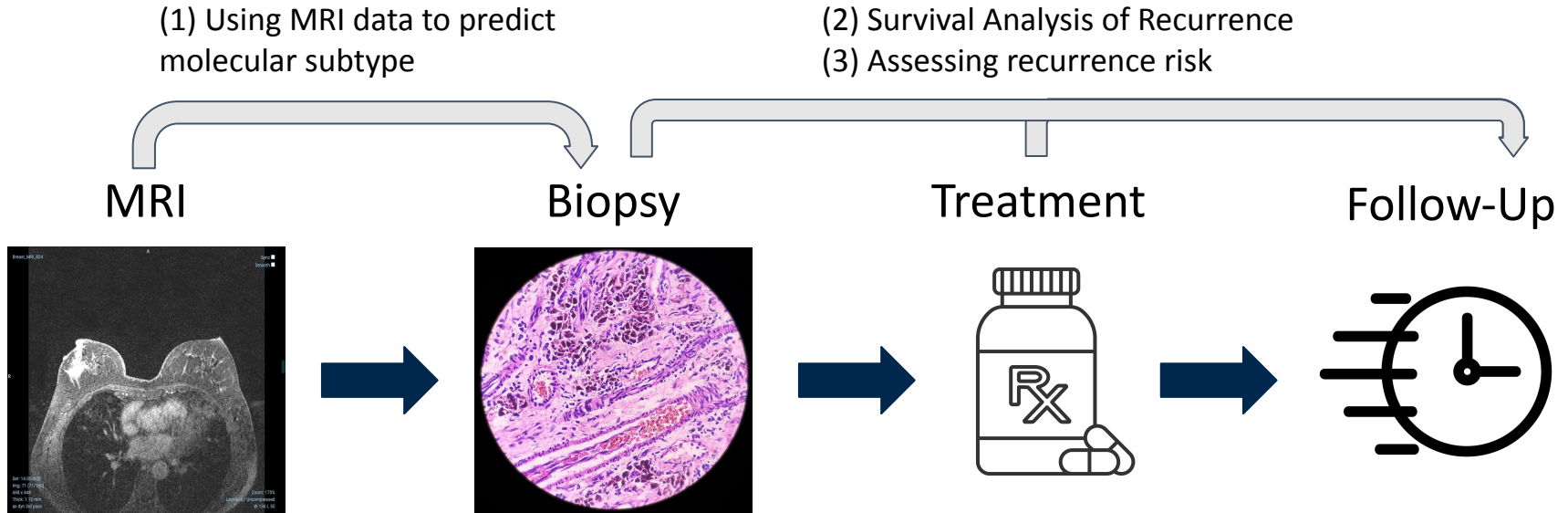
24,240 will **die** from Breast Cancer in 2025

Source: <https://www.nationalbreastcancer.org/breast-cancer-facts/>



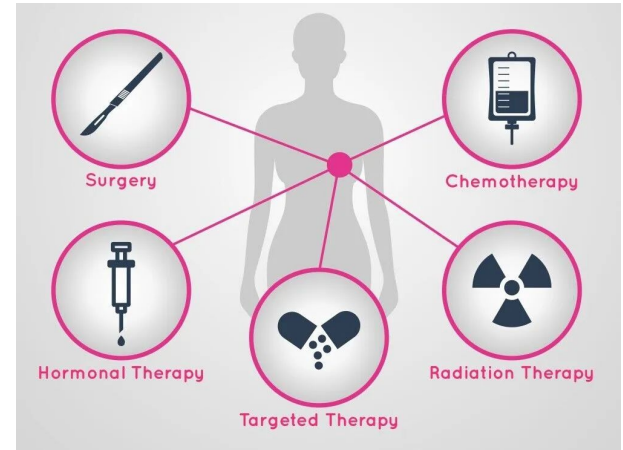
UNIVERSITY OF MICHIGAN HEALTH
MICHIGAN MEDICINE

The Breast Cancer Care Process



Nuances and Acknowledgements

- Our dataset and projects represent only a small snapshot of the journey patients go through.
- Treatment is not a “one size fits all”
 - Neoadjuvant treatments: before surgery
 - Adjuvant treatments: after surgery
 - Many factors go into deciding treatment such as cancer type, clinician expertise, and patient preferences.



Source: <https://nhcancerclinics.com/cancer-types/breast-cancer/>



Background: Dynamic contrast-enhanced MRIs of breast cancer patients

- The Cancer Imaging Archive: Duke Breast Cancer MRI
- Dataset: “single-institutional, retrospective collection of 922 biopsy-confirmed invasive breast cancer patients, over a decade”
- Combination of 96 clinical and 529 imaging features



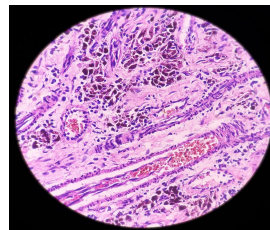
Using radiomics features to predict molecular subtype of breast cancer

Albert Kang, Lucy Malmud, Ritish Natesan, Kate Podrebarac

MRI

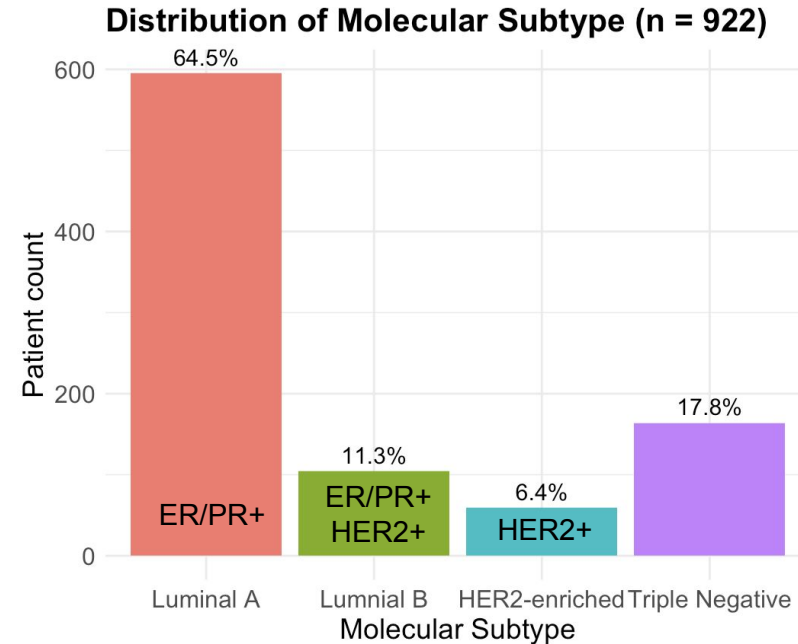


Biopsy



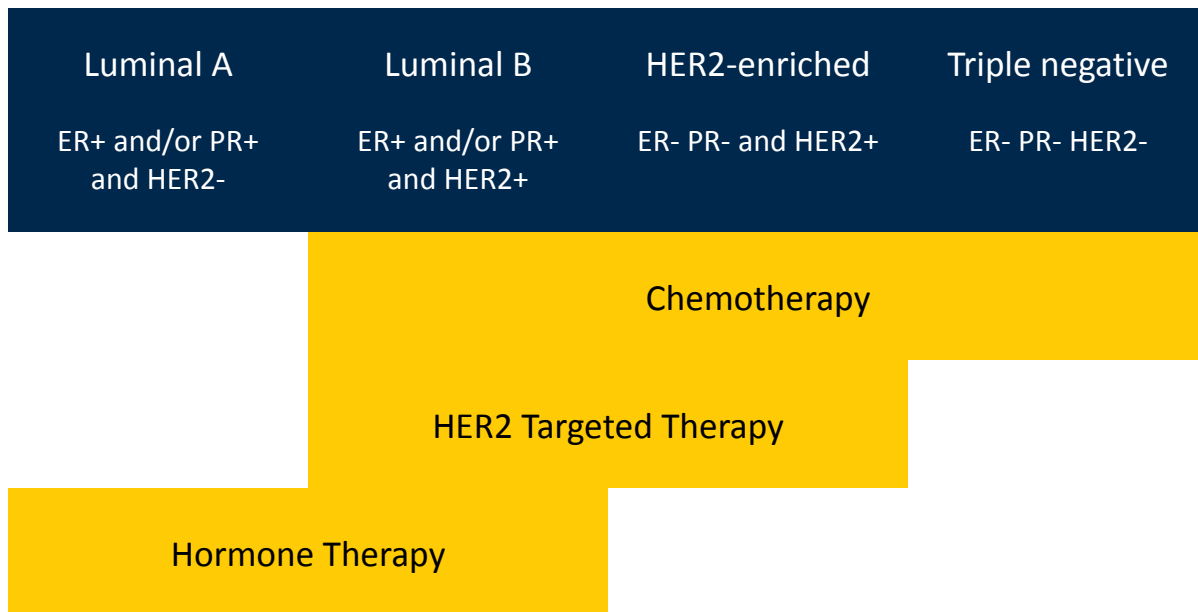
Background: Molecular Subtype

- Combination of Estrogen Receptor (ER), Progesterone Receptor (PR), and HER2 status
- Provides deeper insights into tumor pathology and progression than from histology
- Crucial for accurate diagnosis and precision treatment



Background: Clinical Importance

- Decrease need for invasive biopsy procedures
- Decrease cost of care
- Faster treatment decisions



Background: Reference Paper + Dataset

- Radiomics = the field of study concerned with extracting large amounts of quantitative data from images
 - 529 radiomics features partitioned into 10 feature groups
- In a 2018 study¹ by Saha and colleagues, used random forest model to predict molecular subtype
- Can we do better?

Target	Reference AUC
ER	0.649
PR	0.622
HER2	0.500

¹A Machine Learning Approach to Radiogenomics of Breast Cancer: a study of 922 subjects and 529 DCE-MRI features



Our Process

Full Dataset

Feature Engineering

- PCA
 - Hierarchical clustering
- Pre-biopsy Clinical Variables
- Menopausal status, age, tumor stage, race and ethnicity

16 Feature Subsets

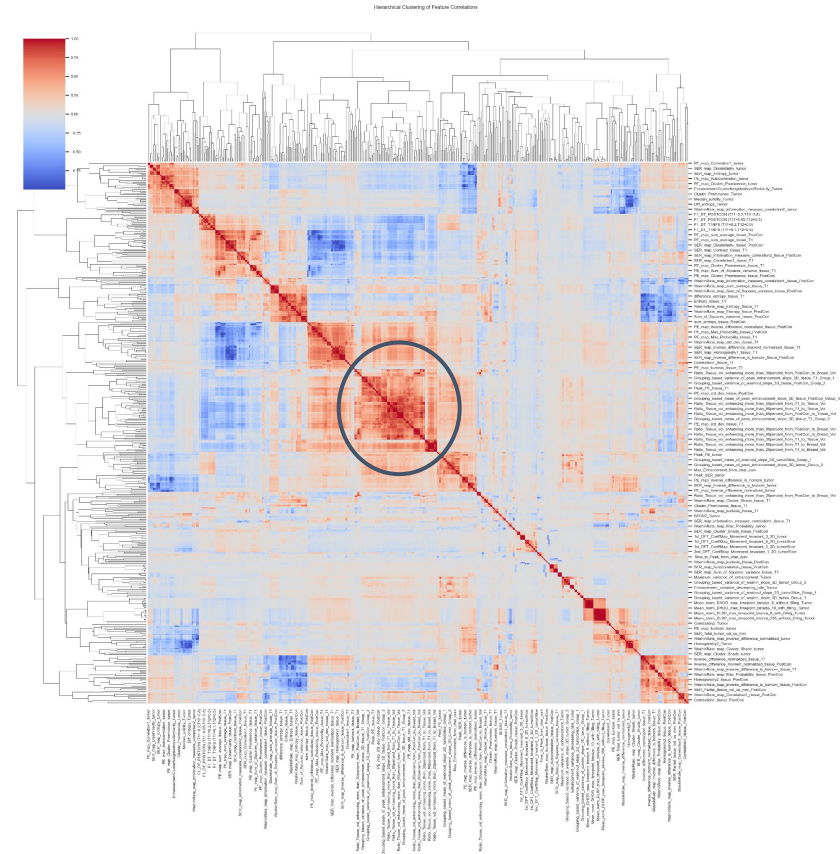
Training: 10 models on each subset

Interpretability

- Permutation Importance

Feature Engineering: Hierarchical Clustering

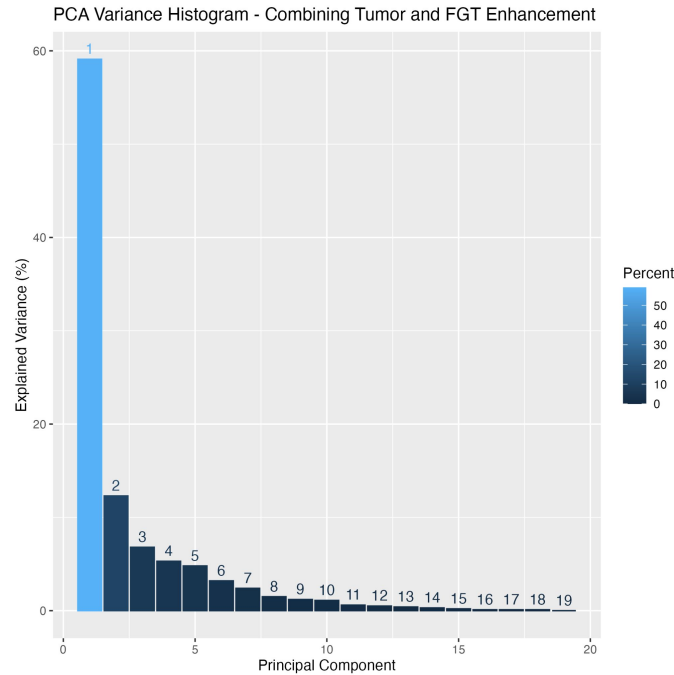
- Correlation-based hierarchical clustering
- “Slice” the dendrogram at a level corresponding to correlation $> .90$
- Select one feature from each of these groups
- 529 $\rightarrow$ 251 covariates



Feature Engineering:

Principal Component Analysis (PCA)

- Transforms high dimensional data into a new coordinate system
- First principal component explains the most variation in the data
- Successive components explain less variation
- Each covariate has one loading factor per component
 - Represents how much that covariate contributes to the component



Training: Models

Generalized Linear Models:

- Logistic Regression
- LASSO
- Elastic Net

Clustering:

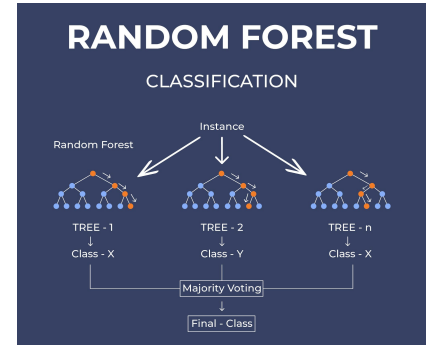
- K-Means
- Gaussian Mixture Model

Machine Learning Models:

- Support Vector Machine (SVM)
- Multilayer Perceptron (MLP)

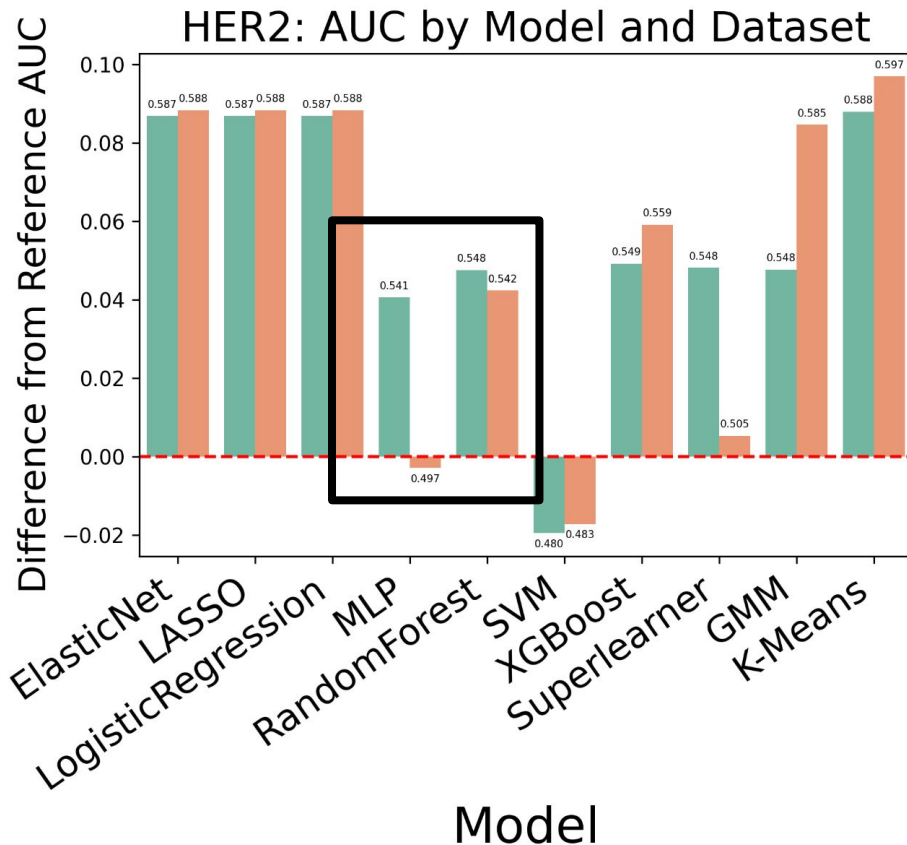
Ensemble Methods:

- Random Forest
- XGBoost
- Superlearner
 - Metalearner: Gradient Boosting Classifier
 - Base Learners: Random Forest, Decision Tree, Logistic Regression, Gradient Boosting Classifier



Source : <https://diagramskiniatinguiccnf.z21.web.core.windows.net/random-forest-model-diagram.html>





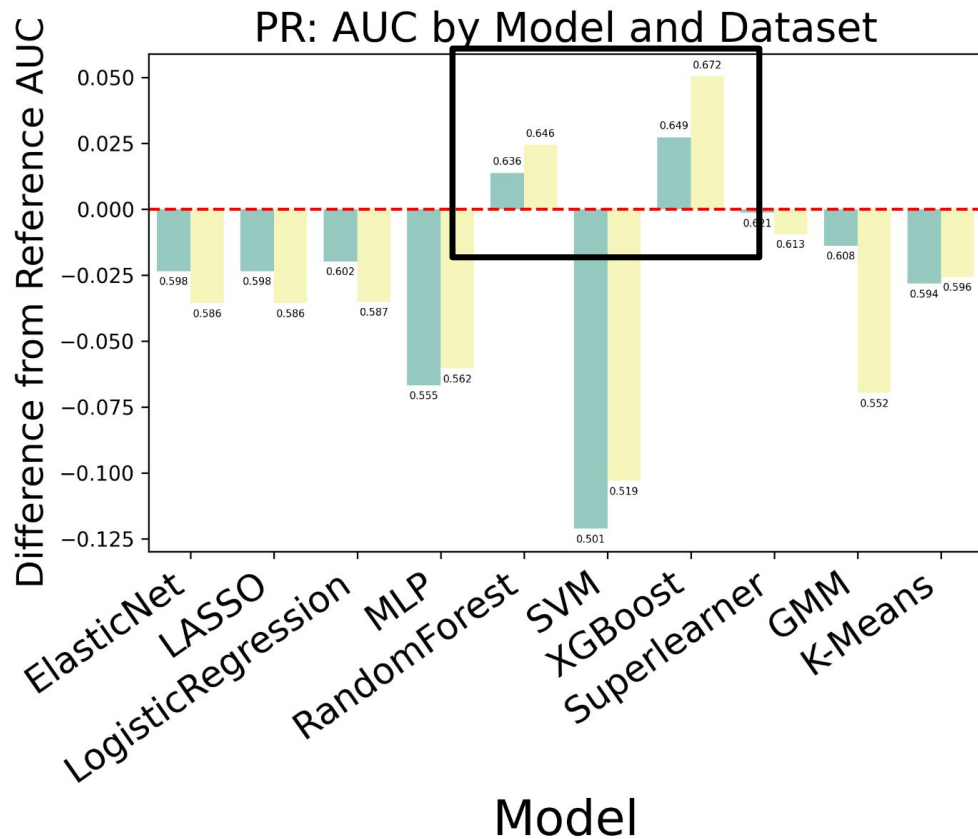
Feature Subsets

- Top Three Principal Components Per Feature Group
- Raw Data Values from Top Three Covariates in PC1

PCA scores yield better results than **raw data values**.

Improvement in performance by capturing more information in the same number of predictors.



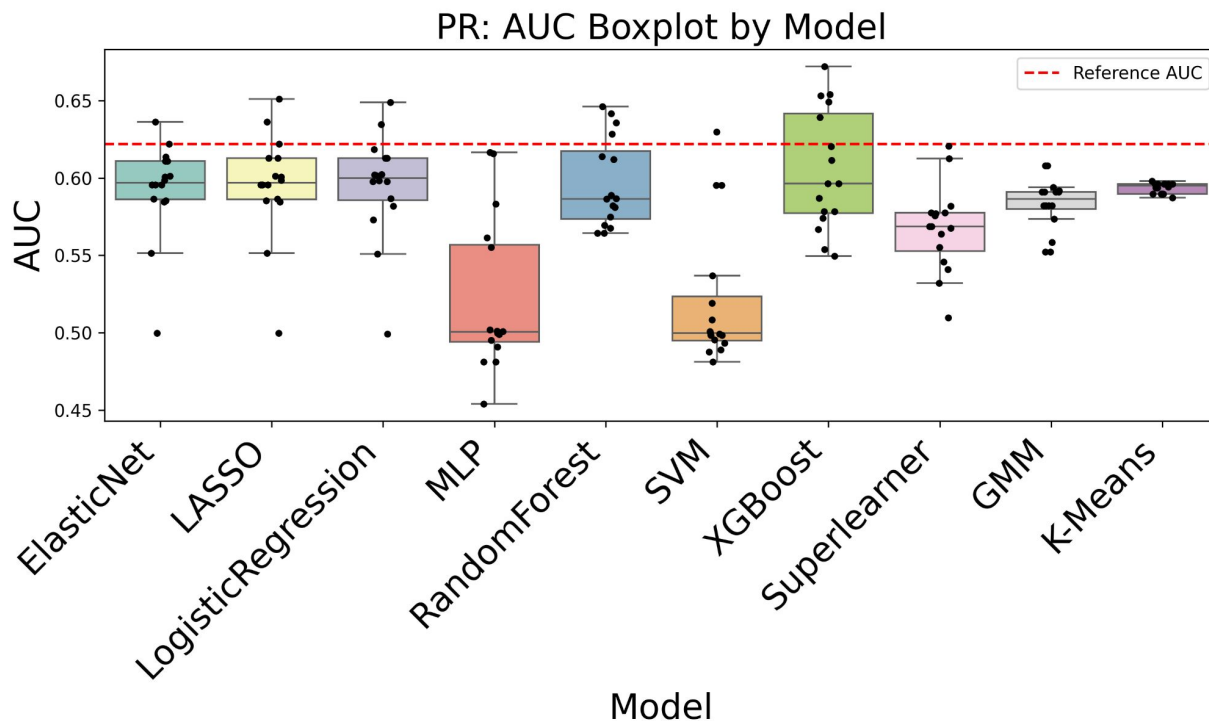


Feature Subsets

- Unengineered Data
- Unengineered Data + Pre-Biopsy Clinical

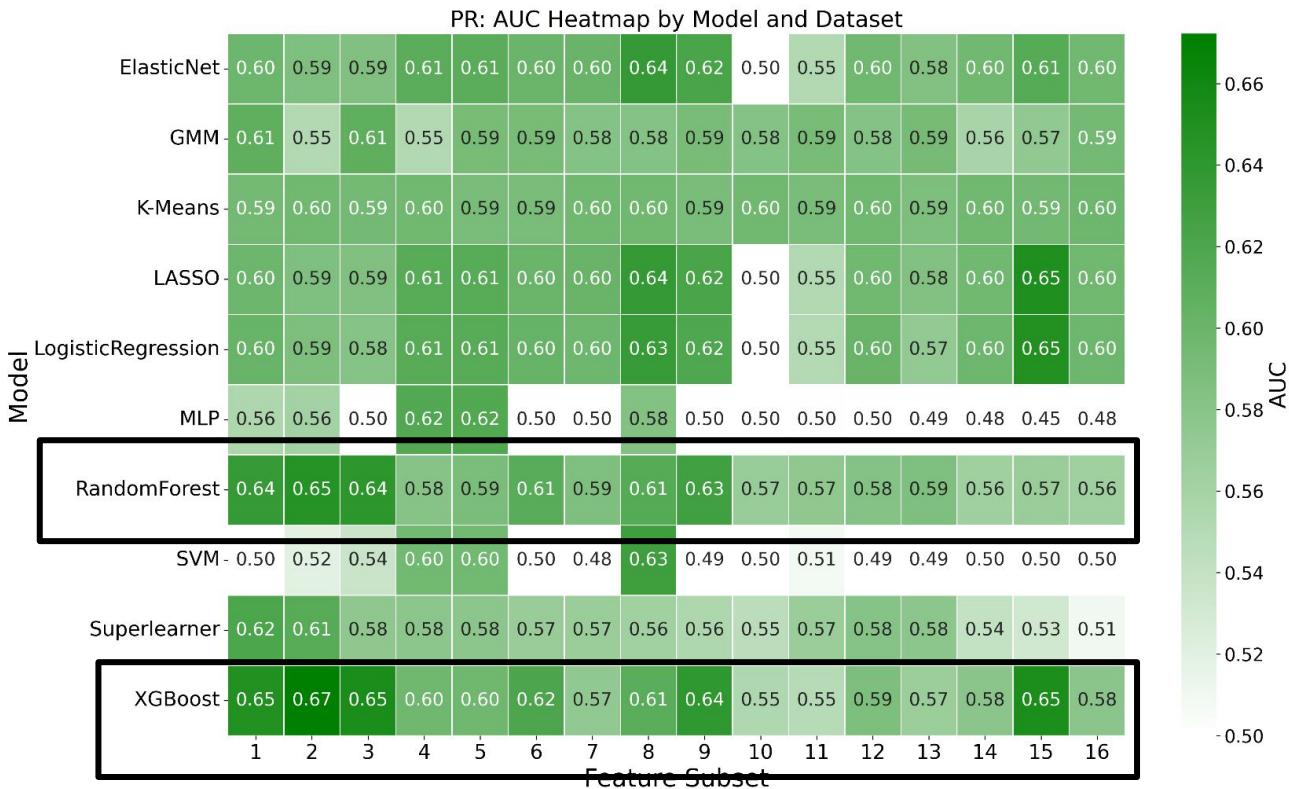
Adding **pre-biopsy clinical features** improves performance as compared to **the original imaging data** in some cases.





- **SVM** and **MLP** consistently perform poorly
- Linear models (**ElasticNet**, **LASSO**, **LogisticRegression**) comparable performance
- **XGBoost**, **RandomForest** consistently best

Feature Subsets



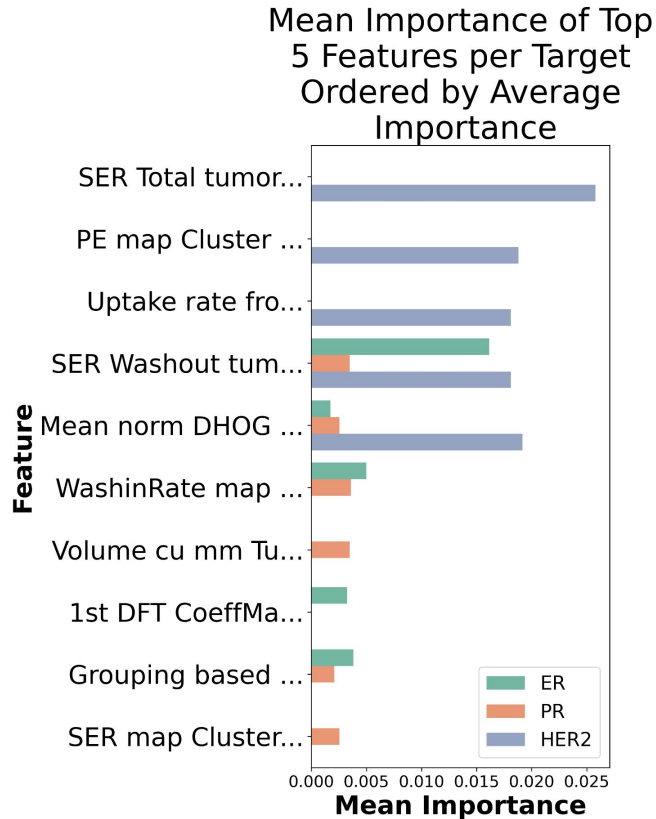
- 1: All Imaging Data
- 2: All Imaging + Pre-Biopsy Clinical
- 3: Hierarchical Clustering
- 4: Hierarchical Clustering + Clinical
- 5: Top PC\ Per Feature Group
- 6: Top Three PCs Per Feature Group
- 7: Top Three PCs s Per Feature Group + Clinical
- 8: Top PC Per Feature Group + Clinical
- 9: Enough PCs to Explain 90% Variance per Feature Group
- 10: Enough PCs to Explain 90% of Variance per Feature Group + Clinical
- 11: Raw Data Values from Top Covariates in PC1
- 12: Raw Data Values from Top Covariate in PC1 + Clinical
- 13: Raw Data Values from Top Three Covariates in PC1
- 14: Raw Data Values from Top Three Covariates in PC1 + Clinical
- 15: Raw Data Values from Enough PCs to Explain 90% of Variance per Feature Group
- 16: Raw Data Values from PCs to Explain 90% of Variance per Feature Group + Clinical

Results

Target	Best Model	Best Dataset	AUC*	Reference AUC
ER	XGBoost	Hierarchical Clustering + Clinical	0.661	0.649
PR	XGBoost	Unengineered + Clinical	0.672	0.622
HER2	Random Forest	Raw Data Values from PCs to Explain 90% of Variance per Feature Group + Clinical	0.655	0.500

Final model and dataset combination with highest *mean AUC across 5-fold cross-validation.

Feature Importance



- Permutation importance measures change in AUC when values of each covariate are permuted
- Insight into what covariates are most important in determining outcome

Important Features

Mean norm DHOG... (3)

- Quantification of texture

SER Washout tumor... (3) + WashinRate map... (2)

- Speed of contrast agent movement

Conclusion



Summary:

- Performing dimensionality reduction was able to increase prediction performance in some settings
- Adding pre-biopsy clinical predictors increased performance
- Training multiple models helped us find the best model for each target

Limitations:

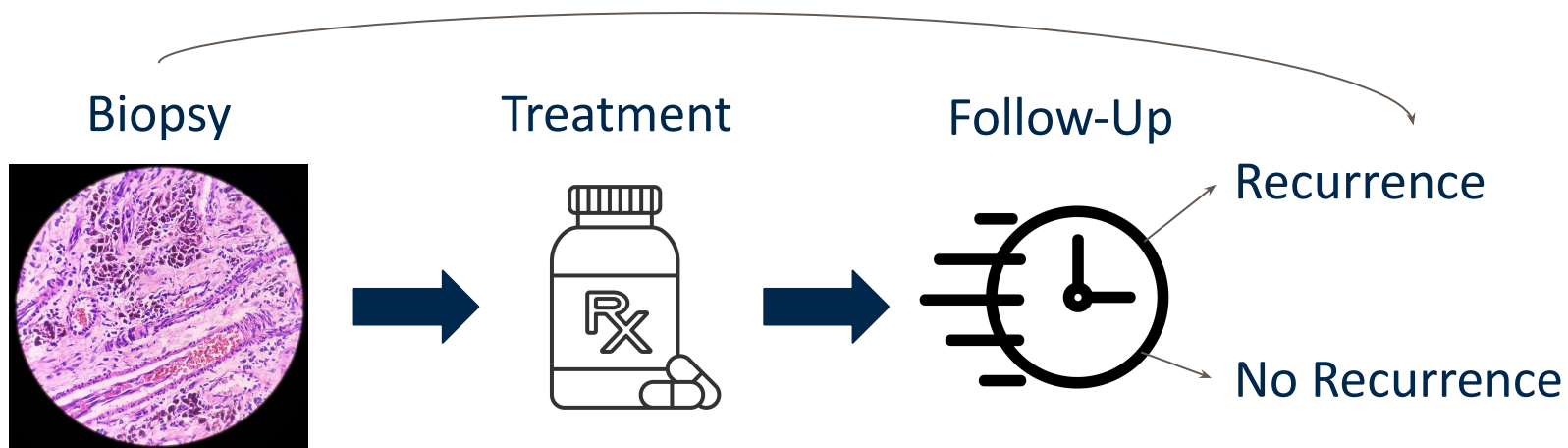
- Imbalanced molecular subtype outcome
- Dataset is restricted to one hospital

Future:

- Using raw MRI data for SVM and MLP

Source: <https://www.svchs.com/blog/southwest-virginia-community-health-systems-recognizes-the-month-of-october-as-breast-cancer-awareness-month/>

Recurrence



Same Dataset

922 Patients with 98 features including:

- Demographics
- Tumor Characteristics
- Mammography & Ultrasound & MRI Findings
- Treatments Types
- Treatment Responses
- Follow-up Information

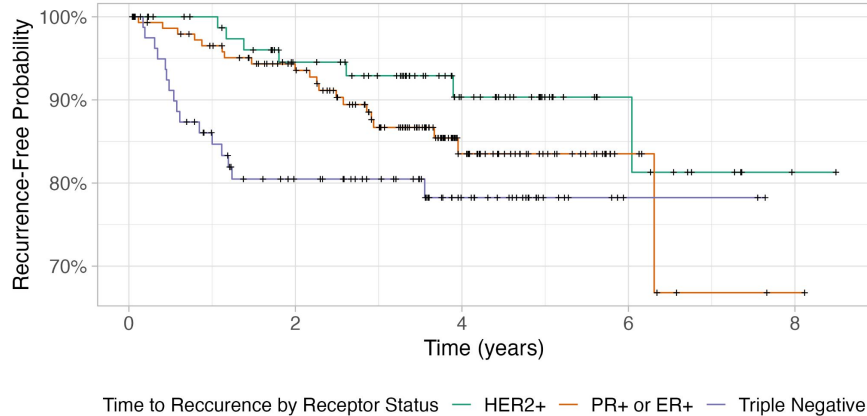
Recurrence among those who underwent **surgery**



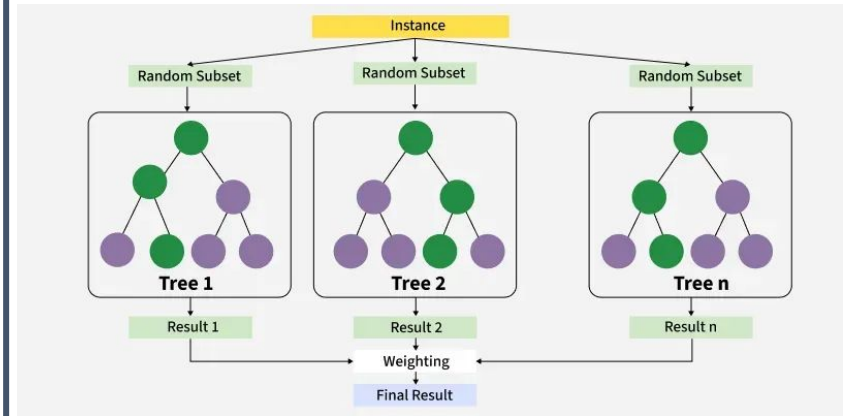
Different Thinking

Survival Analysis

Time to Recurrence by Receptor Status



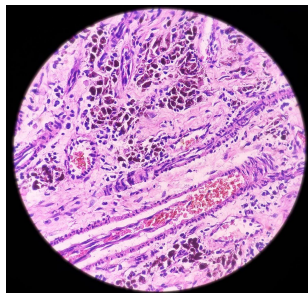
Binary Outcome



Survival Analysis of Recurrence

Carlos Haring & Shiyi Zhang

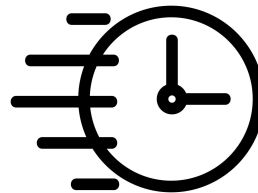
Biopsy



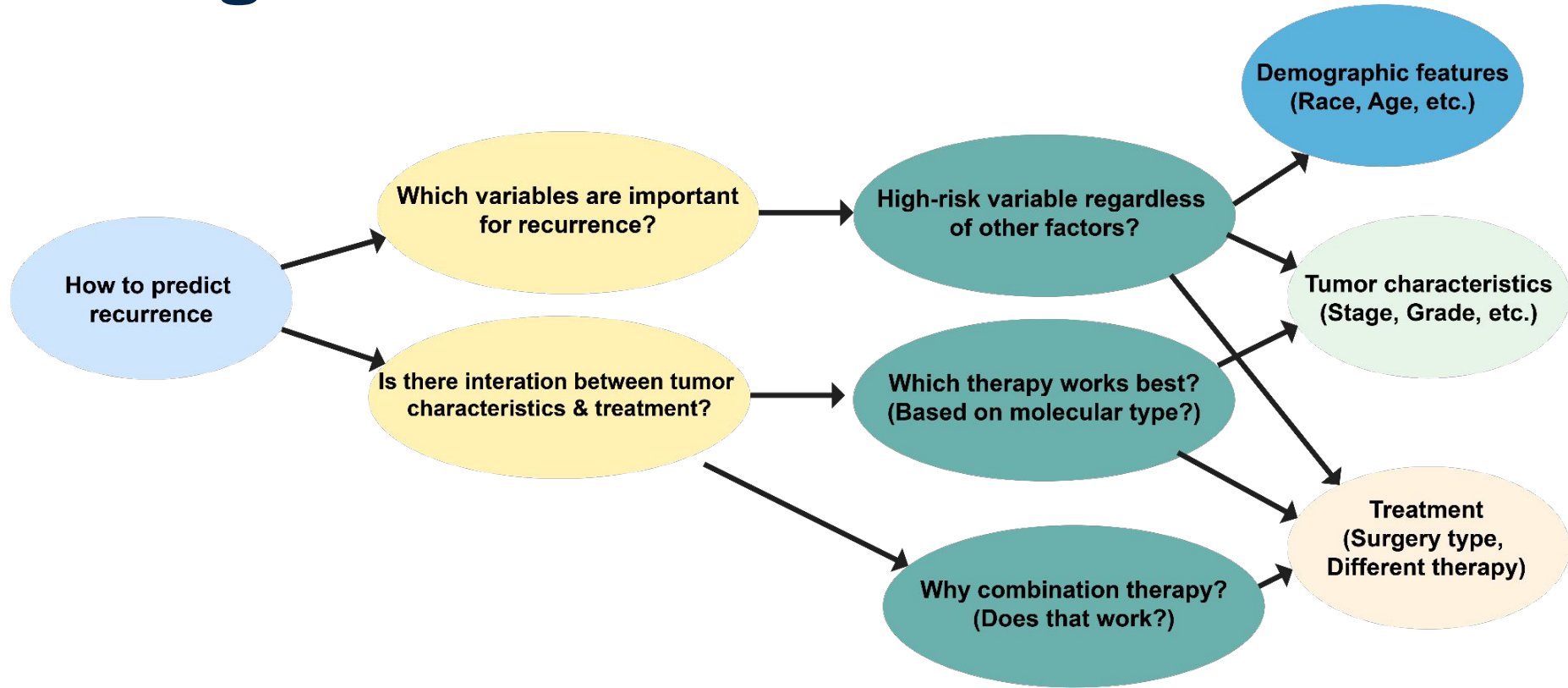
Treatment



Follow-Up



Background and Motivations



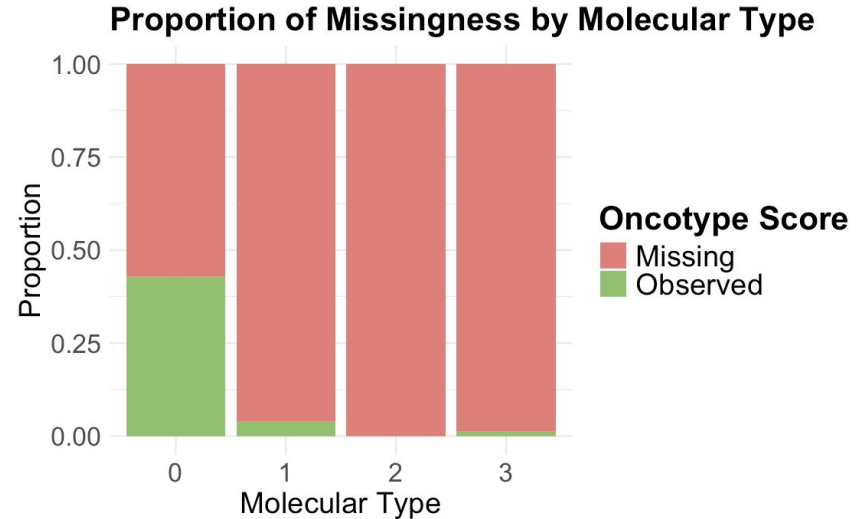
Data Cleaning and Feature Selection

Challenges: Identifying potentially important variables despite:

- **Sparse observations** of events
- **Imbalanced level** in categorical variables
- Data **missing** for clinical reasons

Remedy: Carefully choosing variables by:

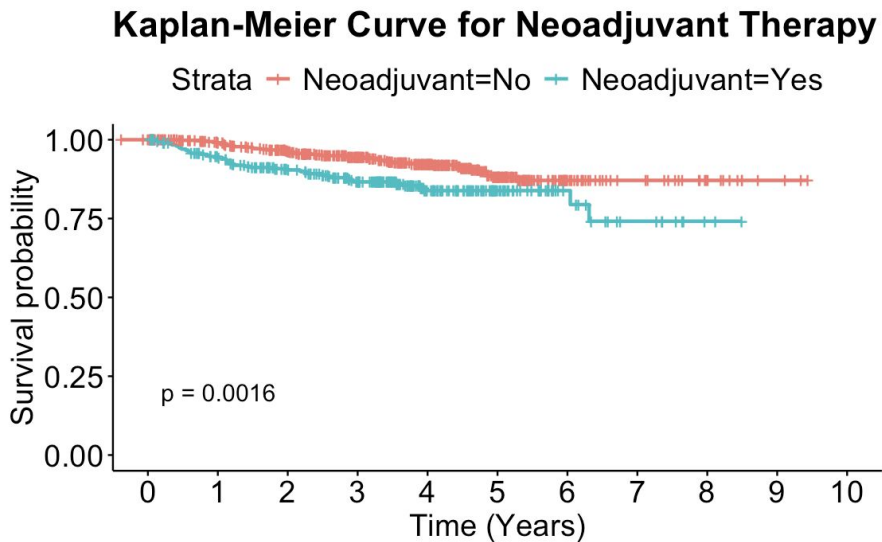
- **Relative significance** in naïve log-rank test
- Omitting **largely unobserved** variables
- **Consolidating** categorical variables levels



Rationale and Methodology

Key Idea: Effectively and efficiently, we need to perform analysis in **subgroup**.

- Only 312 patients underwent **neoadjuvant therapy**;
- **Post-neoadjuvant pathological and clinical measurements** are available only for these patients;
- Such variables are **missing** for the rest of the patients.



Rationale and Methodology

Key Idea: Fit a **stratified Cox Regression** model including variables that are either **clinically meaningful** or significant when run **log-rank test** individually.

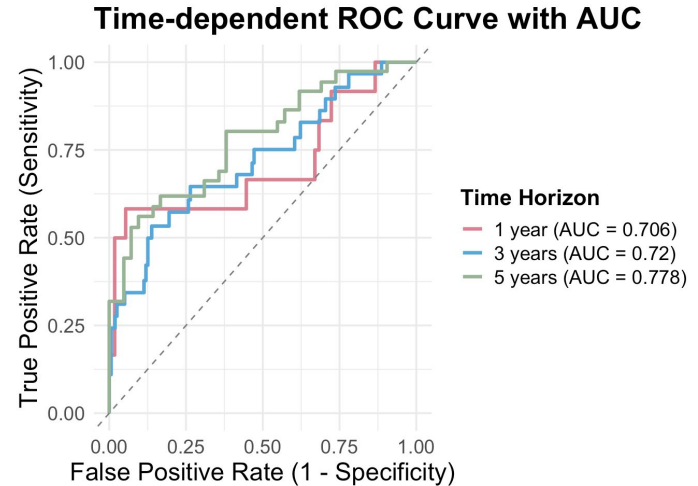
Module	Variables
Demographic	Race
Characteristics at Diagnosis	T stage, N stage, Nottingham score
Treatment	Surgery type, Time to surgery, Chemo, Radiation
Response After Neoadjuvant	Clinical response

Remark: There will be **slight differences** in the selection of variables between the **Cox Regression** and the **Random Forest** model that will be discussed later.



Conclusions and Perspectives

Cox Regression Summary Table				
Rows with different significant level (partial results shown)				
Variable	Reference	Hazard Ratio	95% CI	P-value
T_stage = 2	T_stage = 1	0.590	(0.230, 1.513)	0.272
T_stage = 3	T_stage = 1	0.938	(0.293, 2.998)	0.914
T_stage = 4	T_stage = 1	3.017	(0.817, 11.146)	0.098
N_stage = 1	N_stage = 0	0.515	(0.187, 1.414)	0.198
N_stage = 2	N_stage = 0	2.183	(0.748, 6.369)	0.153
N_stage = 3	N_stage = 0	2.778	(0.916, 8.430)	0.071
Chemo = No Adjuvant	Chemo = Adjuvant	0.857	(0.353, 2.080)	0.734
Radiation = No Adjuvant	Radiation = Adjuvant	0.584	(0.115, 2.957)	0.516
Clinical_response = Complete	Clinical_response = Not Complete	0.268	(0.060, 1.194)	0.084

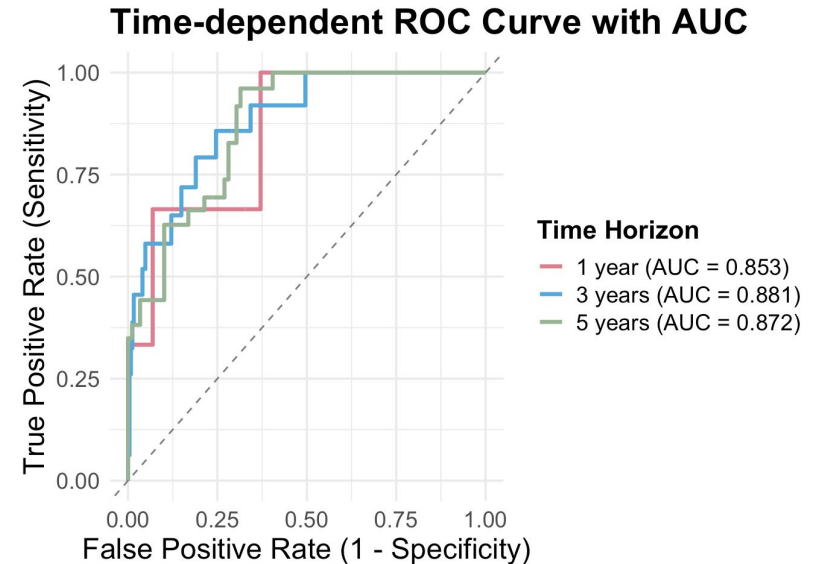


Remark: The **Schoenfeld residuals plot** does not indicate any violation of the proportional hazards assumption. The **concordance index** is 0.727 (SE = 0.051), and the **score test** is statistically significant ($p < 0.001$).

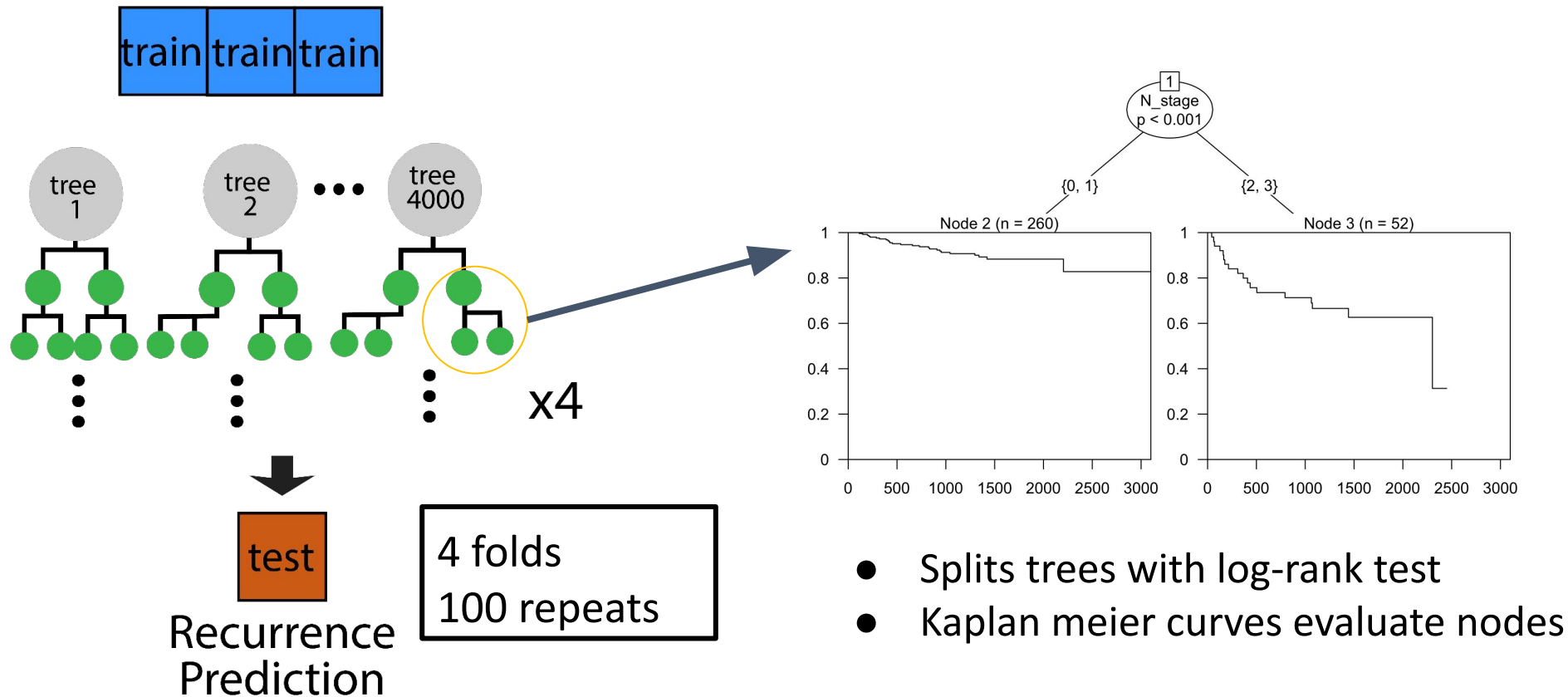
Conclusions and Perspectives

Key Idea: Restrict the analysis to **luminal A** patients **without prior neoadjuvant** treatment to assess the benefit of adjuvant **endocrine** therapy.

Cox Regression Summary Table				
Rows with different significant level				
Variable	Reference	Hazard Ratio	95% CI	P-value
Surgery_type = mastectomy	Surgery_type = BCS	4.057	(1.075, 15.317)	0.039
Time_to_surgery	Continuous	1.008	(1.003, 1.013)	0.001
Nottingham_score	Continuous	1.559	(1.059, 2.296)	0.024
N_stage = 1	N_stage = 0	1.084	(0.304, 3.866)	0.901
N_stage = 2	N_stage = 0	6.607	(1.326, 32.935)	0.021
N_stage = 3	N_stage = 0	1.034	(0.088, 12.121)	0.979
T_stage = 2	T_stage = 1	2.897	(1.013, 8.287)	0.047
T_stage = 3	T_stage = 1	4.263	(0.748, 24.311)	0.103
Chemo = None	Chemo = Adjuvant Only	2.294	(0.779, 6.752)	0.132
Radiation = None	Radiation = Adjuvant Only	1.493	(0.423, 5.272)	0.534
Endocrine = None	Endocrine = Adjuvant Only	4.561	(1.382, 15.050)	0.013

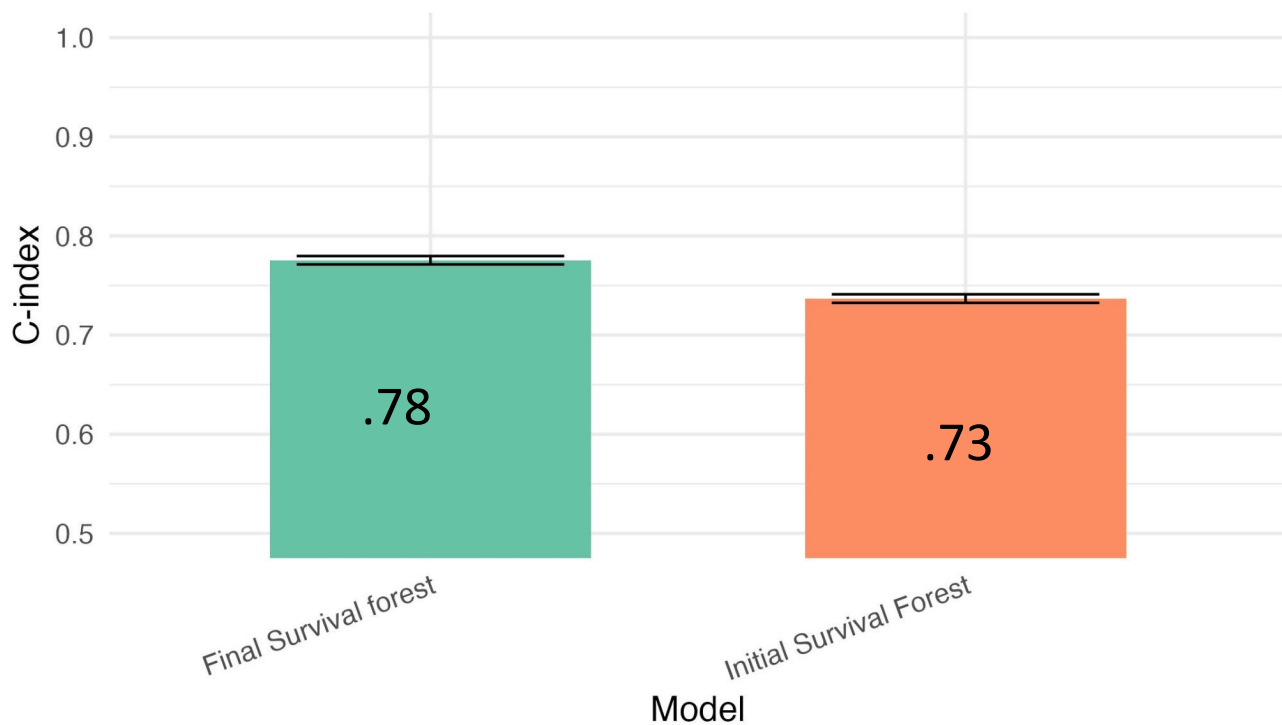


Survival forest with Repeated K-fold cross validation



Model Comparisons

C-index Across Neoadjuvant Models



Initial Survival Forest

- 15 features

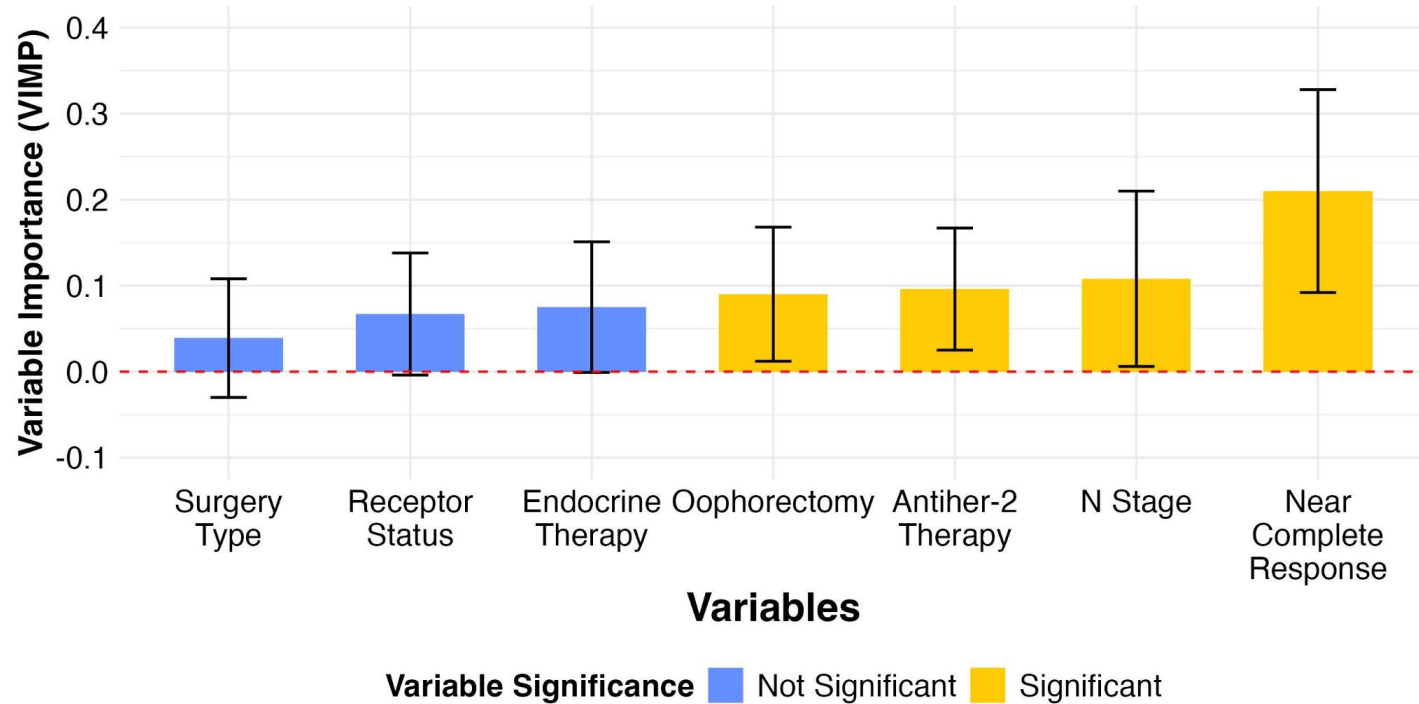
Final Survival Forest

- 7 features

Feature pruning with
Variable importance
(VIMP)

Conclusions and Perspectives

The Importance of Variables Used



Important variables:

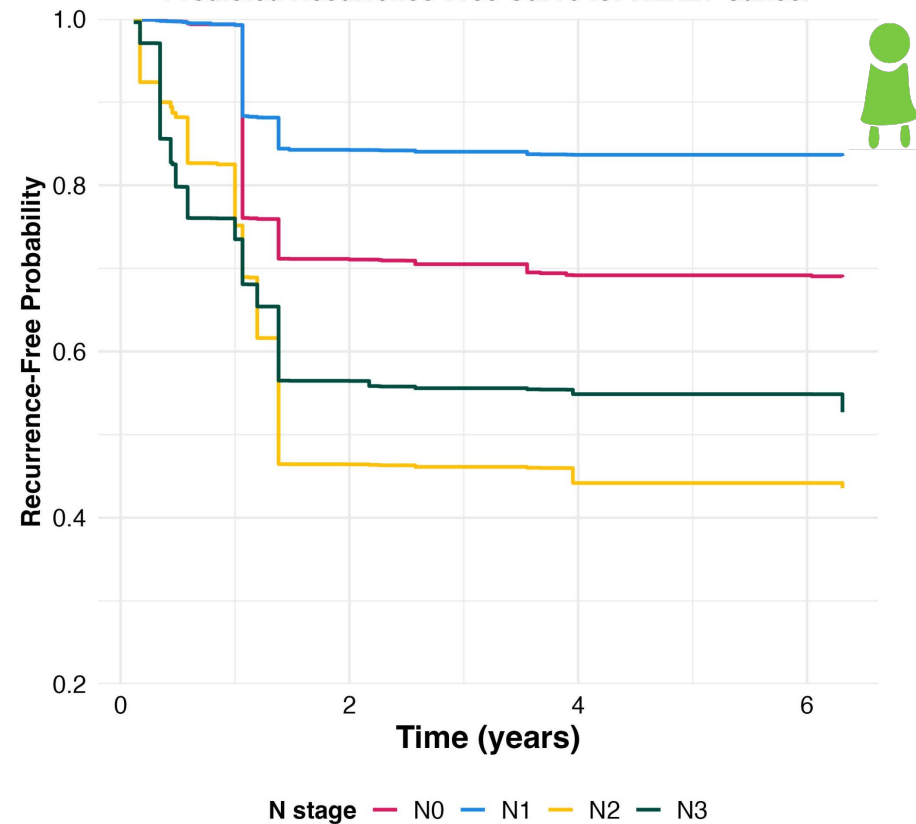
- Neoadjuvant Response
- Treatment Received
- Initial staging

Two scenarios

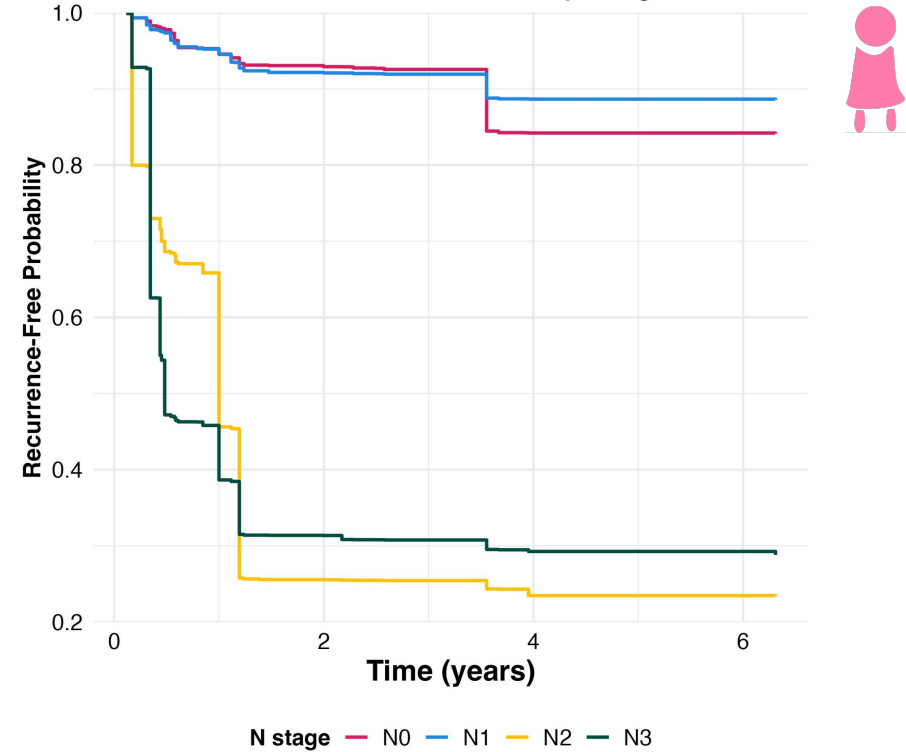
Person	Receptor status	HER2 therapy	Endocrine therapy	Node (N) Stage	Surgery Type	Oophorectomy	Near Complete Response
	Triple Negative	None	None	Vary	BCS	No	Not complete
	HER2+	Neoadjuvant only	None	Vary	BCS	No	Not complete

BCS: Breast Conservation Surgery; HER2+: HER2 receptor positive

Predicted Recurrence-Free Curve for HER2+ Cancer



Predicted Recurrence-Free Curve for Triple Negative Cancer



- Grouping of N0 and N1
- Grouping of N2 and N3
- Triple negative higher risk

Conclusions and Perspectives

What we have done:

- Cox regression offers clinically **interpretable insights** into risk factors;
- Random survival forests achieve **robust** and **high** predictive discrimination.

Future work includes:

- Applying similar methodologies to more clinically meaningful subgroups;
- Utilizing other machine learning methods;
- Investigating from diagnosis to recurrence.

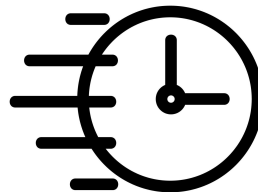
Assessing Recurrence Risk Using Clinical and Molecular Factors

McKenzie Hebert & Sabina Akelbek

Treatment

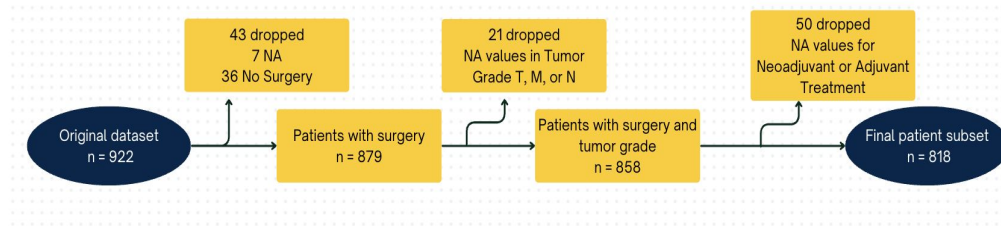


Follow-Up



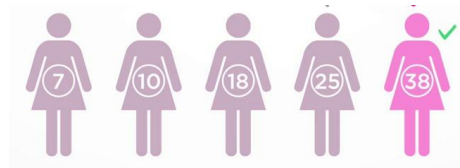
About Our Data:

- Patients Included: 818
 - All patients received surgery
- The features we included in our model:
 - Surgery type (BCS or Mastectomy)
 - Neoadjuvant and Adjuvant Radiation Therapy, Chemotherapy, Endocrine Therapy, & Anti-HER2 Neu Therapy
 - Molecular Subtype
 - Tumor Grades (T, M, and N)
 - Recurrence within 1, 2, and 3 years



Project Goal: Recurrence Risk Probability

- Oncotype Score: A genomic test that predicts recurrence risk and helps guide chemotherapy decisions for patients with early-stage breast cancer

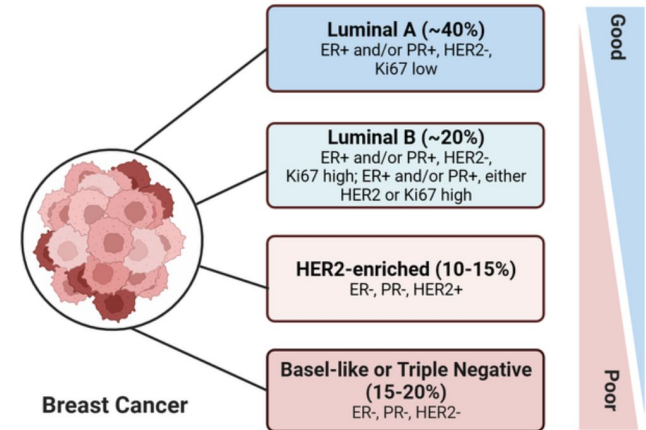


- Our goal is to create a similar system for assessing recurrence risk based on clinical data to inform patient treatment options
- Additionally, we made an interactive web application to assist clinicians in determining a treatment course for patients who are post surgery

Source: <https://www.youtube.com/watch?v=U7iAhzSqf0s>

Project Goal: Clinical Significance

- Only patients with HR+, HER2- breast cancer are eligible to receive an oncotype score
- There may be situations where insurance does not cover all or part of the cost of getting an oncotype score
- The Oncotype Score only predicts the potential benefits of chemotherapy, not including other treatment options



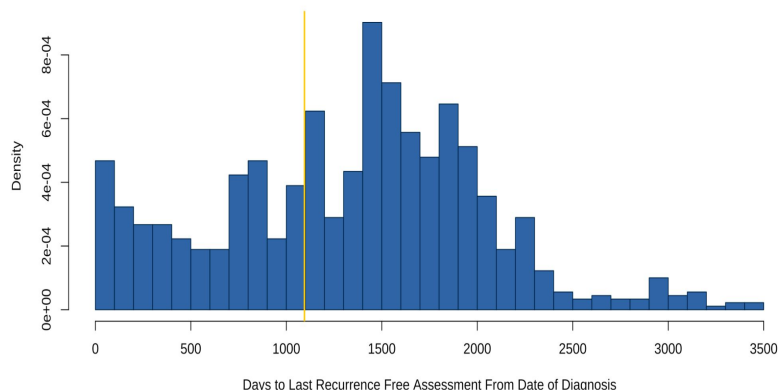
Source: https://www.researchgate.net/figure/An-overview-of-human-breast-cancer-subtypes-This-figure-illustrates-the-various-subtypes_fig1_378762183



Feature Engineering: Recurrence

- Our model treats a recurrence event as a binary outcome
- Minimizing the effects of right censoring
 - Included patients as “non-recurrence” if follow-up time was less than 3 years or recurrence event was after 3 years.

Distribution of Days to Last Recurrence Free Assessment From Date of Diagnosis

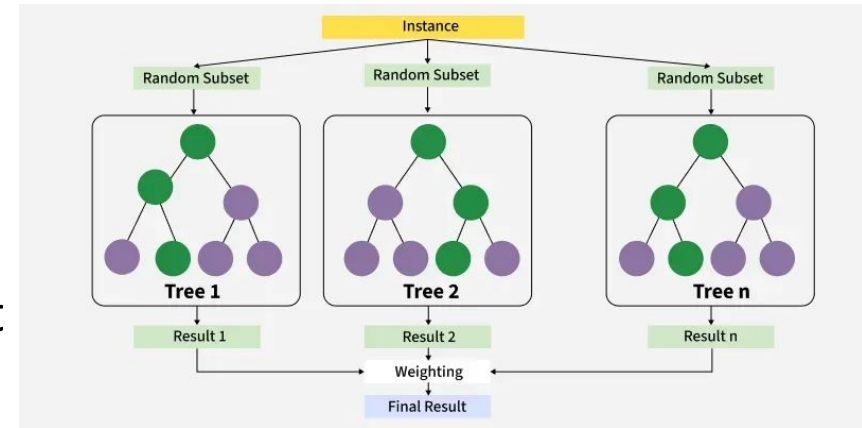


Time	Followed (%)	No Recurrence	Recurrence
1 Yr	91.6%	806	12
2 Yrs	83.5%	780	38
3 Yrs	69.4%	762	56
Total	100%	734	84



Prediction Model: XGBoost

- XGBoost: A machine learning method that builds on prior errors in a series of decision trees to generate a final probability score
- Works very well on highly imbalanced datasets
- Allows us to view feature importance
- Built-in LASSO regularization to prevent overfitting

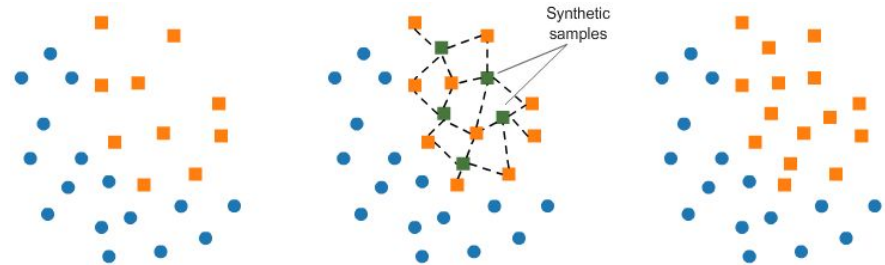


Source: <https://www.geeksforgeeks.org/machine-learning/xgboost/>



Prediction Model: XGBoost

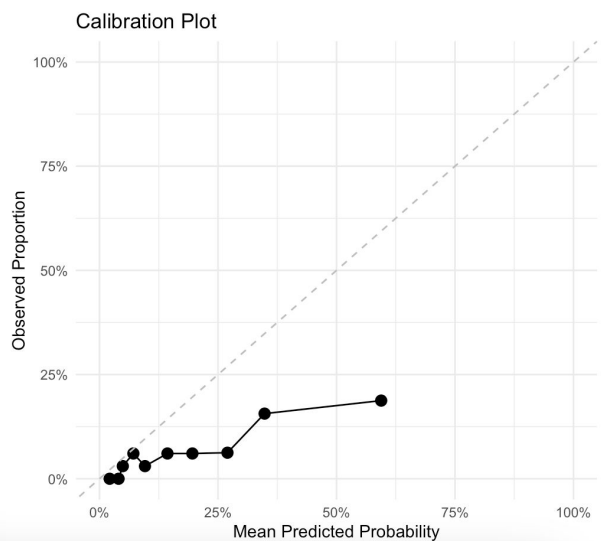
- XGBoost has a built in parameter for addressing imbalance data
 - Class Weighting Method
- **SMOTE: Synthetic Minority Oversampling Technique**
 - Creates synthetic samples of the minority class to balance the dataset and improve model performance
- Hyperparameter Tuning
 - GridSearch with Cross Validation



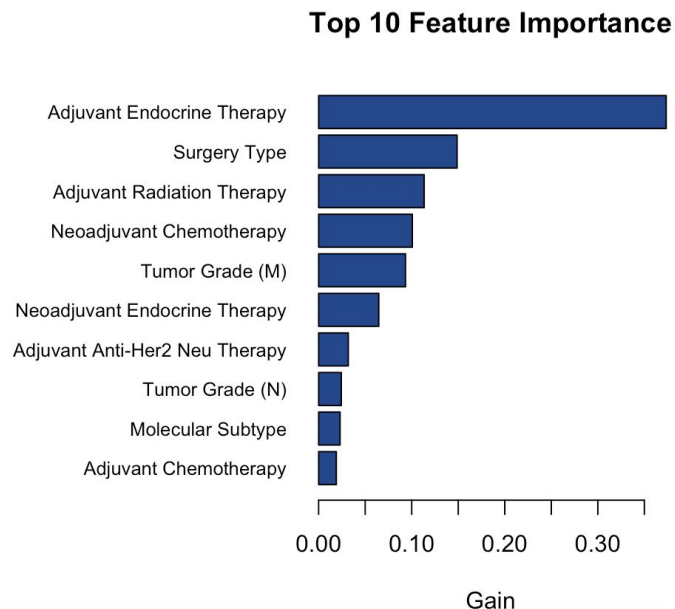
Source: <https://medium.com/@asheshdas.ds/oversampling-to-remove-class-imbalance-using-smote-94d5648e7d35>

Model Evaluation

- Evaluation Metrics:
 - AUC: 0.75
 - Brier Score: 0.07

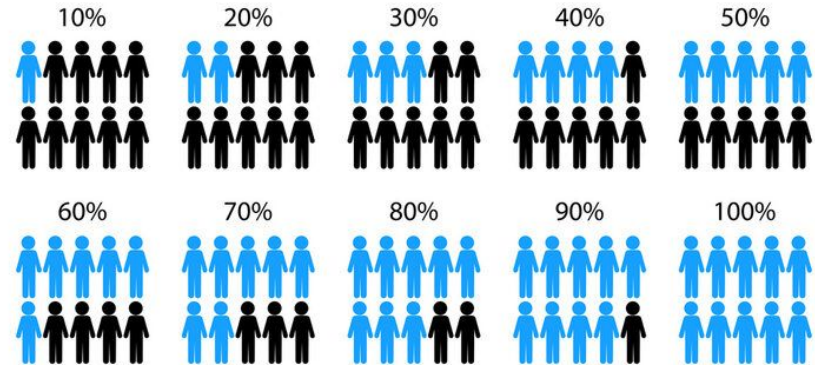


- Feature Importance:



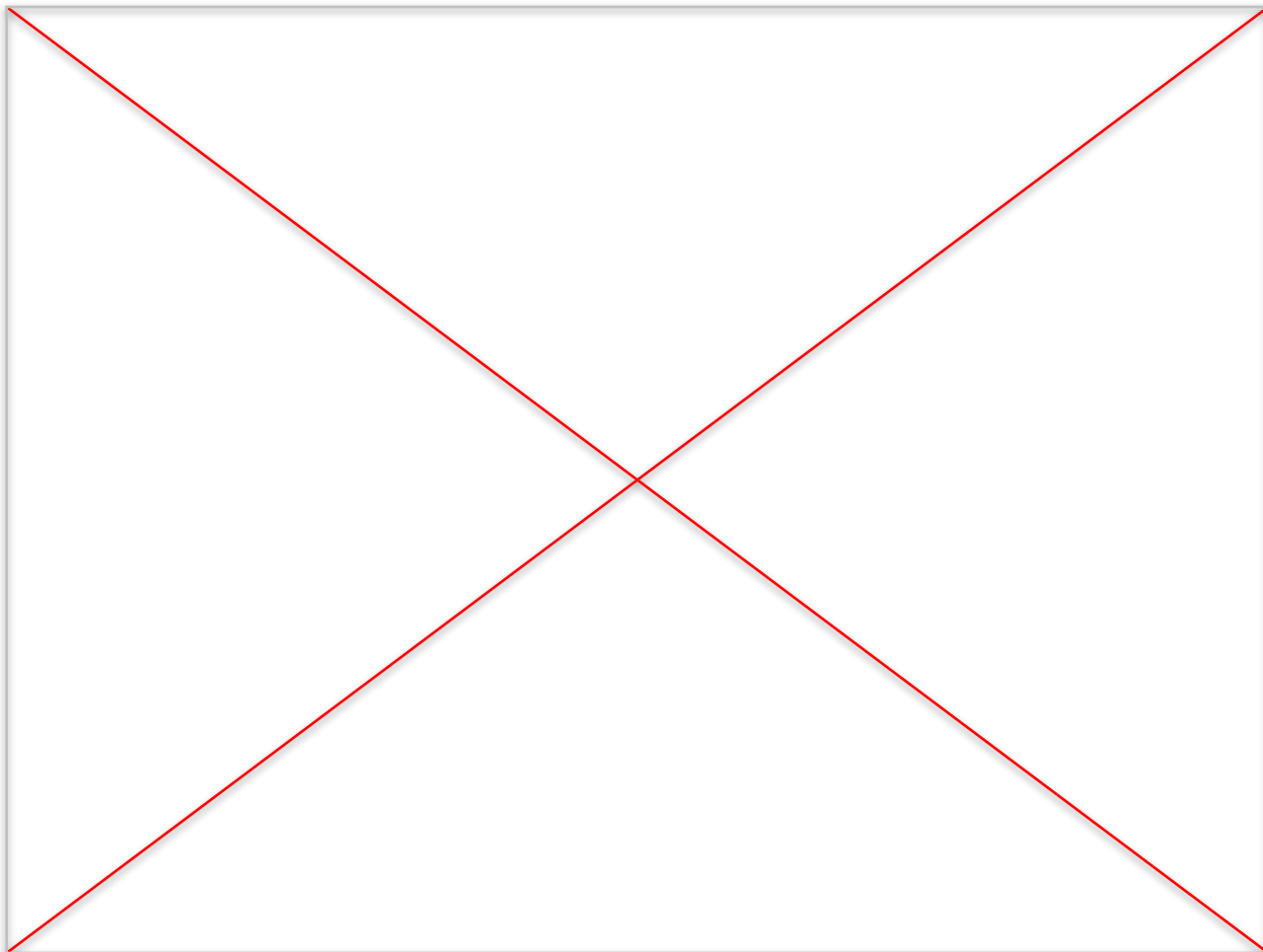
Web Application using R Shiny

- We made an interactive app where clinicians can input their patients' diagnosis data and neoadjuvant treatments.
 - Clinicians can use the score to determine adjuvant treatment course
 - Patients can use the score to visualize their risk
- Human pictogram display to make the score easily digestible to both patients and clinicians.



Source: <https://medium.com/@asheshdas.ds/oversampling-to-remove-class-imbalance-using-smote-94d5648e7d35>





Breast Cancer Recurrence Risk Predictor

Our prognostic model is designed for clinicians assisting patients diagnosed with breast cancer. It's intended for use after patients have completed neoadjuvant therapy and surgery. This tool helps predict the likelihood of cancer recurrence within three years, providing valuable insight for selecting appropriate adjuvant therapy options.

Enter your information:

Patient information (not used in prediction):

Age:

43

Surgery type:

☐ Breast-Conserving Surgery (BCS)

☒ Mastectomy

Select all treatments before surgery:

Chemotherapy

☒ No ☐ Yes

Radiation Therapy

☒ No ☐ Yes

Endocrine Therapy Medications

☒ No ☐ Yes

Anti-HER2/Neu Therapy

☒ No ☐ Yes

Select all treatments after surgery:

Chemotherapy

☒ No ☐ Yes

Radiation Therapy

☐ No ☒ Yes

Endocrine Therapy Medications

☐ No ☒ Yes

Your Predicted Probability of Recurrence (3 years):

16.5%

Out of 100 people like you, 16 may have a recurrence



Limitations



- Model uses binary recurrence (yes/no)
- Predictions are limited to a 3-year timeline and only 69.4% of the patients were followed up with to then.
- For clinical use, other predictors such as sociodemographic and comorbidity data should be incorporated.

Source: <https://www.svchs.com/blog/southwest-virginia-community-health-systems-recognizes-the-month-of-october-as-breast-cancer-awareness-month/>

Future Work



- Repeat this project with a different approach
 - Time-to-Event Prediction
 - Other clinical features not available to us
- Expand web application functionality to provide more detailed information about risk
- Comparing our model to oncotype score predictability in a prospective trial.

Source: <https://www.svchs.com/blog/southwest-virginia-community-health-systems-recognizes-the-month-of-october-as-breast-cancer-awareness-month/>

Thank You!